



New York Times Predictor

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Link : [NYT Bestseller Predictor](#)

[API](#)

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Project Overview & Motivation

The publishing industry spends ~\$29.9B trying to answer how likely a book can make the NYT Bestseller List.

Motivated by independent authors, small publishers, literary agents



Key Objective:

This project builds a machine learning model using gradient boosting to predict a book's likelihood of appearing on the New York Times Best Seller list. Using features drawn from publisher reputation, author's history, release timing, publication year, page count, and genre.

The model identifies patterns that distinguish bestsellers from comparable titles.



The goal is to give publishers and authors a data-driven tool to evaluate a manuscript's commercial potential early in the publication pipeline.

Data Collection and Preprocessing

NYT Books API: title, author, publisher, rank, list name, ISBN, and weeks on list

Google Books API: publication date, page count, description, categories, publisher, language, and maturity rating

Open Library API: edition count, subjects, ebook access, languages, first publication year, author information, and page counts.

- matched using ISBN-13 digit value so records could be merged across APIs
- negative sampler: helped model the difference between bestsellers and non-bestsellers

Preprocessing

Positive NYT bestsellers	1,811
Negative non-bestsellers	5,493
Training Rows	5,843
Testing Rows	1,461
Training class ratio	4,394 : 1,449 3 : 1

Final Dataset Characteristics

Clean ISBN-13
Title - All
Author - All
Publisher - NYT/GG
Publish year - GG
Genre - GG
Page count - OL
Edition Count - OL
Subjects - OL
Ebook Access - OL
Languages - OL
First Publish Year - OL
Is a Bestseller

Challenges encountered and solutions

- Matching book records
- Missing metadata
 - had to pull OL page number
- Class Imbalance
 - Randomize the collection and pulled more weeks for NYT bestsellers
 - The goal is to predict whether a book is likely to become the bestseller

Model Development and Evaluation

Feature engineering

19 model features

Numeric / Derived Features

- publish year, page count, *edition count, first publish year, book age, *first publish age, publication gap, subjects, *languages

Missingness / Coverage Indicators

- page count, edition count, subjects, ebook access, languages, first publish year

Categorical / Encoded Features

- *publisher grouped, author grouped, *ebook access, language, *subjects

Model Development

Trained & Tested:

Baseline and Tree-based models

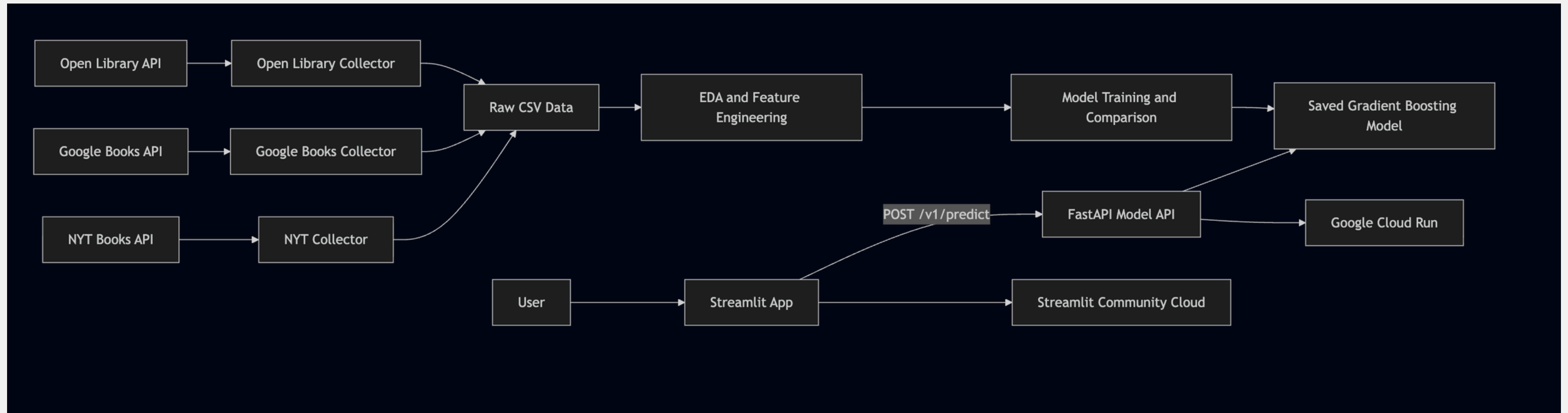
Model	AVG Precision in test set
Logistic Regression	0.9615
Random Forest	0.9618
*Gradient Boosting	0.9637
XGBoost	0.9630

Evaluation (metrics)

*Gradient Boosting Model

Average precision	0.9637
ROC AUC	0.9788
F1 score	0.9094
Recall	0.8453
Accuracy	0.9582

Solution Architecture



Frontend: Streamlit Cloud

Backend API: FASTAPI on Cloud Run

Container: Docker

Model: scikit-learn Gradient Boosting

Data/EDA: pandas, matplotlib, seaborn

Deployment: Cloud Build + Artifact Registry

Application

NYT Bestseller Predictor

Estimate the likelihood that a book could become a New York Times bestseller.

How to use the predictor

Enter as much book information as you know then submit the form to estimate bestseller likelihood. Blank fields are allowed, but fuller information usually gives the model a stronger signal. The result is a probability estimate, not a guarantee of future bestseller status.

Book Information

Title
Spring Forward

Author
Jasmin Pecina

Publisher
Simon & Schuster

Original publication year ⓘ Edition publication year ⓘ
2019 2020

Page count ⓘ Number of editions ⓘ
275 1 - +

Genres or subjects ⓘ
Fantasy, Romance

Language ⓘ
eng

Digital availability ⓘ
No digital copy available ▾

Predict Bestseller Likelihood

Prediction

Enter book information and submit the form to get a prediction.

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Predict Bestseller Likelihood

Prediction

Bestseller Probability

59.5%

Likely Bestseller

Model details

```
{  
  "model_name" : "gradient_boosting"  
  "threshold" : 0.5  
  "prediction" : 1  
}
```

App Walkthrough:

User enters information → User clicks “Predict Bestseller Likelihood” → Streamlit app sends the data to the FastAPI endpoint → API loads the trained model and returns a probability → App displays the bestseller likelihood and prediction label.



Challenges Faced and Lessons Learned

Challenges Faced

- Book metadata from external APIs can be inconsistent.
 - Number of page numbers → Had to compare google books api and open library api
 - Some ISBNs return incomplete or mismatched records.
- Negative Samples in modeling → Had to collect more NYT data to compare to negative samples and clean

Role of AI assistants in development

- Coding Challenges
- Helped create the FastAPI prediction API
- Helped design feature engineering from the available metadata
- Helped build foundational Streamlit user interface
- Architecture Diagram

Lessons Learned / Improvements

- Cleaning the dataset before modeling can be cumbersome
- Data quality and feature design are just as important as model choice
- Deployment has its own challenges separate from modeling
 - A model that works locally still needs an API, containerization, environment variables, and a frontend that can call the deployed service correctly.
- Time constraint (fine-tuning, larger & diverse dataset)
- Expand the dataset across more years and bestseller categories.
- Add more features like sales data
- Address overfitting